

Marine A. Denolle

Associate Professor, Earth and Space Sciences, University of Washington

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Research Impact Summary

Research Citations: 3,398 total (h-index: 24, i10: 45) | 2,291 last-5y (h₅: 22, i10₅: 43) | Google Scholar
Publications: 66 peer-reviewed | 3 Nature/Science-family (T1) | 40 AGU-flagship (T2) | 17 domain journals (T3) | 3 reviews/perspectives | 3 preprints
Research Funding: Lead PI: \$3.1M (14 grants) | Co-PI/Co-I: \$4.9M (8 grants) | Fellowships: \$878K (2)
Open-Source Software: 372 GitHub stars | 241 forks | 18 active repos | NoisePy: 207 stars, 83 forks, 19 contributors | 1,692+ PyPI downloads/yr
Mentoring: 8 PhD (4 current, 4 graduated) | 11 postdocs | 15+ undergrads | 10 co-supervised grad students | 5 became faculty
Broader Impact: 6 workshops, 215+ seismologists trained | mlgeo-book: 11 GitHub stars

Education

2008 – 2014 **Stanford University** Geophysics
 • Supervisor: Dr. Gregory Beroza
 • Co-supervisors: Dr. Eric Dunham, Dr. Jesse Lawrence
 2007 – 2008 **Ecole Normale Supérieure - IPGP** Geophysics
 • Supervisors: Dr. Satish Singh (IPGP), Dr. David Bercovici (Yale)
 2006 **Ecole Normale Supérieure** Earth Sciences
 2004 – 2005 **Lycée Chateaubriand, Rennes** Physics-Mathematics

Employment

Applied Environmental Intelligence
 Seattle
 2025 – present
Co-Founder
 University of Washington
 Seattle, WA
 2024 – present
Associate Professor
 Department of Earth and Space Sciences
 University of Washington
 Seattle, WA
 2021 – 2024
Assistant Professor
 Department of Earth and Space Sciences
 Harvard University
 Cambridge, MA
 2016 – 2021
Assistant Professor
 Department of Earth and Planetary Sciences
 UC San Diego, Scripps Institution of Oceanography
 La Jolla, CA
 2014 – 2016
Green Postdoctoral Fellow
 Institute of Geophysics and Planetary Physics
 • Supervisor Dr. Peter Shearer

Awards

- 2023 **Invited Professorship** Ecole Normale Supérieure rue d'Ulm-Paris
- 2023 **Data Science Fellow** eScience Institute, University of Washington
- 2019 **Charles F. Richter Early Career Award** Seismological Society of America
- 2019 **Kavli Frontiers of Science Fellow** National Academy of Sciences
- 2019 **Radcliffe Assistant Professorship** Institute for Advanced Study
- 2018 **CAREER Award** NSF
- 2017 **David and Lucile Packard Foundation Fellowship** David and Lucile Packard Foundation
- 2016 **Outstanding Reviewer** Geophysical Research Letters
- 2015 **Outstanding Reviewer** Geophysical Journal International
- 2012 **AGU Outstanding Student Paper Award** American Geophysical Union
- 2012 **SSA Student Presentation Award** Seismological Society of America
- 2010 **AGU Outstanding Student Paper Award** American Geophysical Union

Teaching

- ESS 469/569 **Machine Learning in the Geosciences**
U of Washington geo-smart.github.io/mlgeo-book
2021 - 2024
- Taught Fall 2021, 2022, 2023, 2024
 - Open-access Jupyter Book with asynchronous teaching materials
 - Course adopted by University of Arizona and UC Berkeley
 - Curated geoscience-specific ML datasets for homework
- ESS 563 **Computational/Advanced Seismology**
U of Washington denolle-lab.github.io/teaching/ess563
2023 - 2025
- Taught Spring 2023, 2024, 2025
 - Covers ambient noise seismology, earthquake source characterization
 - Julia and Python programming for high-performance computing
- ESS 412/512 **Introduction to Seismology**
U of Washington denolle-lab.github.io/teaching/ess412
2023 - 2026
- Taught Winter 2023, 2025, 2026
 - Covers earthquake physics, seismic waves, and hazard assessment
- ESS 314 **Geophysics**
U of Washington github.com/UW-geophysics-edu/ESS314-fall23
2022 - 2026
- Taught 2022, 2023, 2024, 2026
- EPS 268 **Machine Learning in Earth and Planetary Sciences**
Harvard
2019
- Taught Fall 2019

- EPS 268 **Induced Seismicity**
Harvard
2018 • Taught Fall 2018
- EPS 203 **Earthquakes and Faulting**
Harvard
2018 - 2020 • Taught Fall 2018, 2020
- EPS 55 **Earthquakes and Tectonics**
Harvard
2017 - 2020 • Taught Fall 2017, 2020
- EPS 204 **Earthquake Sources**
Harvard
2016 • Taught Spring 2016

Phd Advisees

- PhD Pre-Candidate **Michael Hemmett**
UW ESS
2025 - present Offshore geophysics
- PhD Candidate **Manuela Koepfli**
USS
2022 - present Geohazard assessment and early warning systems
• 1 publication, 1 in review, 1 in prep
- PhD Pre-Candidate **Akash Kharita**
UW ESS
2022 - present Geohazard characterization using machine learning
• 2 publications
- PhD Candidate **Yiyu Ni**
UW ESS
2021 - present Machine learning and big data seismology
• 8 publications
- PhD **Congcong Yuan**
Harvard
2019 - 2024 Time-dependent seismology, solid-fluid interaction. Next position: Cornell postdoc → Faculty position at NTU Singapore
• 4 publications
- PhD **Stephanie Olinger**
Harvard
2018 - 2023 Cryo-seismology (50% co-advised with Brad Lipovsky). Next position: Stanford Thompson Postdoctoral Fellowship → CEO of AEI
• 4 publications
- PhD **Tim Clements**
Harvard
2016 - 2021 Hydro-seismology and big-data seismology. Next position: USGS Mendenhall Postdoc → USGS Research Geophysicist
• 4 publications
- PhD **Jiuxun Yin**
Harvard
2016 - 2022 Earthquake seismology and source characterization. Next position: Caltech SCSN Postdoc Fellowship → Schlumberger
• 6 publications

Postdocs

- University of Washington **Dr. Qibin Shi**
2022 - 2024 DAS, Deep Learning, Earthquake Catalog building, AgroSeismology
• Now Postdoctoral Fellow at Rice University
- University of Washington **Dr. Kuan-Fu Feng**
2023 - 2024 Environmental Seismology, Cloud computing
• Next position: Postdoc at University of Utah
- University of Washington **Dr. Ethan Williams**
2023 - 2025 DAS
• Next position: Assistant Professor at UC Santa Cruz
- University of Washington **Dr. Stephanie Olinger**
2024 - 2025 Cryoseismology, DAS, Ambient noise monitoring
• Next position: Climate Tech → CEO Applied Environmental Intelligence
- Harvard University/University of Washington **Dr. Laura Ermert**
2020 - 2022 Environmental seismology
• Next position: Assistant Professor at ISTerre
- Harvard University **Dr. Xiaotao Yang**
2019 - 2020 Ambient noise cross correlation
• Next position: Assistant Professor at Purdue
- Harvard University **Dr. Kurama Okubo**
2019 - 2020 Ambient noise monitoring
• Next position: Researcher at NIED, Japan
- Harvard University **Dr. Zhitu Ma**
2019 - 2020 Ambient noise cross correlation
• Next position: Assistant Professor at Tongji University, China
- Harvard University **Dr. Chengxin Jiang**
2018 - 2019 Ambient noise seismology, computational seismology
• Next position: Research Associate at Australian National University
- Harvard University **Dr. Chris Van Houtte**
2016 - 2017 Earthquake Source seismology
- Harvard University **Dr. Loïc Viens**
2016 - 2018 ambient noise seismology, ground motion prediction
• Next position Researcher at Los Alamos

Other Graduate Student Supervision

Note: (*) At UW, roles are secondary advisor or primary advisor on one project/manuscript. (**) Advising resulted in a peer-reviewed publication.

Maleen Kidiwela (*, *): 2021-present, UW Oceanography (co-advised with William Wilcock) - DAS and offshore seismology, resulted in publication

Zoe Krauss (*, *): 2021-2024, UW Oceanography (co-advised with William Wilcock) - Axial Seamount hydrothermal monitoring, resulted in publication

Parker Sprinkle (*, *): 2021-present, UW ESS (co-advised) - Earthquake early warning and machine learning

Natasha Toghramadjian (*, *): 2018-2025, Harvard EPS - Seismic attenuation and site effects, resulted in publications

Zhuo Yang (*, *): 2018-2021, Harvard EPS - Machine learning for seismic phase picking, resulted in publication

William Flanagan (*, *): 2019, Harvard EPS - Volcano seismology project

Thibault Pérol (*, *): 2017, Harvard EPS - Deep learning for earthquake detection, resulted in multiple publications

Congcong Yuan (*, *): 2019, USTC China - Master student visiting researcher on time-dependent seismology, resulted in publication

Philippe Danré (*, *): 2018, Ecole Normale Supérieure Paris - Master thesis on seismic monitoring, resulted in publication

Manuel Florez (*, *): 2017-2019, MIT - Dissertation committee member for geophysics PhD

Graduate Student Committees

Koepfli, Manuela: Earth and Space Sciences, Chair, 2024-present

Ni, Yiyu: Earth and Space Sciences, Chair, 2024-present

Kharita, Akash: Earth and Space Sciences, Advisor, 2023-present

DeGrande, Jensen: Earth and Space Sciences, Member, 2023-present

Pearson, Anna Elaine Rogers: Earth and Space Sciences, Member, 2024-present

Sprinkle, Parker: Earth and Space Sciences, Member, 2022-present

Kidiwela, Maleen: School of Oceanography, Member, 2024-present

Krauss, Zoe: School of Oceanography, GSR, 2022-2024

Sangmin, Song: School of Oceanography, GSR, 2024-present

Zhang, Maochuan: School of Oceanography, GSR, 2023-present

Jones, Randall: Atmospheric Sciences, GSR, 2025-present

Chien, Mu-Ting: Atmospheric Sciences, GSR, 2023-2024

Sweeney, Aodhan: Atmospheric Sciences, GSR, 2024-present

Ragland, John: Electrical and Computer Engineering, GSR, 2023-2024

Rasanen, Ryan: Civil and Environmental Engineering, GSR, 2022-2023

Velappan, Hemalatha: School of Environmental and Forest Sciences, GSR, 2023-present

Velegar, Meghana S: Applied Mathematics, GSR, 2023-2023

Zahn, Olivia: Physics, GSR, 2022-2024

International Phd Advising

Note: Dissertation and defense evaluating committee members

Marius Paul Isken : 2024 (GFZ, Germany)

Luc Illien : 2023 (GFZ, Germany)

Zoe Renate : 2023 (Universite de Lorraine, France)

Reza Esfahani : 2022 (GFZ, Germany)

Kurama Okubo : 2019 (ENS, Paris, France)

Media Coverage

- [2025 Global-scale Database of Seismic Phases](#)
- [2024 Ambient Field Monitoring in the Shallow Crust](#)
- [2024 Making Big Data Look Small](#)
- [2023 UW eScience Seminar](#)
- [2023 Open Science in Seismology](#)
- [2023 UW News Fiber Sensing](#)
- [2018 Radcliffe Seminar: Where we stand in Earthquake Prediction](#)

Undergraduate Students

Note: My undergraduate advising includes research opportunities through summer programs and independent studies (ESS 499 for credit during academic year, hourly pay during summer/breaks). I mentor students toward conference presentations and publications, and support competitive fellowship applications. Legend: (*) Conference presentation | (**) Peer-reviewed publication | (***) Publication in prep | (+) NSF GRFP recipient

Alex Rose : 2025-present, UW Oceanography - DAS data processing research

Anjani Mirchandi: 2024-present, UW Applied Math - DAS data processing research

Hiroto Bito: 2023-2026, UW ESS - ML for earthquake catalog building and spatio-temporal forecasting

Nicholas Wolfe: 2023, UW ESS - Earthquake magnitude estimation project

Informatics Capstone Team (lead Rona Guo(*)): 2023, Nathan Limono, William Phan, Michael Yung, Matthew Herradura, UW Informatics - DAS Platform development, resulted in conference presentation

Lucas Swanson : 2023, UW Informatics - DAS software development

Francesca Skene (*) : 2022-2023, UW ESS - Surface event catalog, resulted in conference presentation

Nick Smoczyk (*,)** : 2022-2024, University of Minnesota - Volcano seismology using RedPy catalog, resulted in conference presentation and publication in prep

Julian Schmitt (*,+) : 2020-2021, Harvard EPS - Ambient noise seismology on cloud computing with Julia, BASIN project, resulted in conference presentation and NSF GRFP

Jared Bryan (*,,+)**: 2019, Harvard EPS - Ambient noise monitoring, SCEC Internship, resulted in conference presentation, publication, and NSF GRFP

Albert Aguilar (*): 2018, Harvard EPS - Subduction zone seismology and data mining, SCEC, resulted in conference presentation

Leore Lavin: 2016, Harvard EPS - Ground motion simulations with ambient noise

Roy Bowling (*): 2014, Scripps IGPP - Ambient noise seismology, SCEC Internship

Tara Larrue (*): 2012, Stanford Geophysics - Ambient noise seismology, SURGE program

Penprapa Wutthijuk (*): 2011, Stanford - Ambient noise seismology, SURGE program

Training Workshops

2025 (2025).

2024 (2024).
2024 (2024).
2023 (2023).
2021 – present (2021 – present).
2016 (2016).

University Service

University of Washington **Department Colloquium Committee**
2026 – present • Help organizing the department colloquium series

University of Washington **AI@UW Advisory Committee to the VP**
2026 – present • Advising AI strategy and initiatives across the university, commitment ~ 1hr/week

University of Washington **Co-director of UW CS4Env**
2024 – present • Computer Science for the Environment (CS4Env) is a cross-unit initiative to promote synergies between computer science and environmental research at UW, includes organizing biweekly seminar, reviewing proposals, and organizing an annual symposium.

UW/ESS **Chair of Curriculum Committee**
2024 – present • Reviewing proposed courses, course changes, undergraduate and graduate program.

UW/ESS **Member of Curriculum Committee and Data Science Oversight Committee**
2022 – present • Review the proposed data science option program

UW/ESS **Member of Executive Committee**
2022 – 2023

UW/ESS **Member of Research Faculty Search Committees**
2022 – 2023 • Search committee for PNSN Network Manager and Igneous Processes Faculty Search

Harvard University **Multiple Committee Memberships**
2016 – 2020 • Undergraduate Curriculum Committee, Graduate Student Council, IT Committee, Diversity Inclusion and Belonging Committee, Department Colloquium Committee

Professional Service

2026 – present **Member of Board of Directors at Earthscope Consortium**

2023 – 2025 **Chair of Integration and Innovation Advisory Committee at Earthscope Consortium**

2024 – 2025 **Member of Science Steering Committee at SCEC**

2021 – 2022 **Member of Data Service Standing Committee at IRIS**

2021 – 2023 **Member of HPC Standing Committee at SCEC**

2011 – 2012 **Chair of Graduate Student Council at Stanford University**

2022 – 2022 **Committee Member of Charles Richter Early Career Award Committee at Seismological Society of America**

Review Panels

2020,2024 **NSF Review Panels EAR-Geophysics and GeoInformatics**
• Reviewed 10+ proposals per panel

2016 – 2018 **USGS Review Panels Earthquake Hazards Program**
• Reviewed 10+ proposals per panel

Editorial Service

- 2025 – present **Editor at Geophysical Journal International**
- 2017 – 2020 **Associate Editor at Geophysical Research Letters**
- 2014 – present **Reviewer for multiple journals in geophysics and seismology**
- >150 reviews for GJI, BSSA, Nature Communications, GRL, JGR, Science, and others

Publications Legend

Author Notation: **Bold:** Marine A. Denolle (PI). *Italic:* graduate students, postdocs, and undergraduate students mentored by MD.

Publications

- 2026 Veronica Gaete-Elgueta, *Manuela Köpfli*, Dominik Gräff, Bradley P Lipovsky, **Marine A. Denolle**, Weston A. Thelen, Brendan Pratt, Taylor R. Kenyon (2026). **Distributed Acoustic Sensing Records of Earthquakes and Surface Processes at Mount Rainier Volcano.** in review in *Seismica*
- 2026 John Townend, Ilma Del Carmen Juarez-Garfias, Olivia Pita-Sllim, Calum J. Chamberlain, Emily Warren-Smith, Caroline Holden, Kasper van Wijk, **Marine Denolle**, Andrew Curtis, Hiroe Miyake (2026). **Seismological Analysis of Contemporary and Future Alpine Fault Earthquakes Using the Southern Alps Long Skinny Array (SALSA).** in press in *Seismological Research Letters*
- 2026 *Natasha Toghramadjian*, **Marine A. Denolle**, *Laura Ermert*, *Chengxin Jiang* (2026). **Probing the Seattle Basin Edge Using a Dense Urban Nodal Array in 100 Backyards.** *Seismological Research Letters*, [10.1785/0220250241](https://doi.org/10.1785/0220250241)
- 2026 Alexey Yermakov, Yue Zhao, **Marine Denolle**, *Yiyu Ni*, Philippe M. Wyder, Judah Goldfeder, Stefano Riva, *Jan Williams*, David Zoro, Amy Sara Rude, Matteo Tomasetto, Joe Germany, Joseph Bakarji, Georg Maierhofer, Miles Cranmer, J. Nathan Kutz (2026). **The Seismic Wavefield Common Task Framework.** arXiv, [10.48550/arXiv.2512.19927](https://arxiv.org/abs/10.48550/arXiv.2512.19927)
- 2026 *Qibin Shi*, David Montgomery, Abigail L.S. Swann, Nicoleta C. Cristea, *Ethan Williams*, Nan You, Joe Collins, Ana Prada Barrio, Simon Jeffrey, Ana Misiewicz, Tarje Nissen-Meyer, **Marine A. Denolle** (2026). **Agroseismology: unraveling the impact of farming practices on soil hydrodynamics.** in review in *Science* - preprint on Arxiv, [10.48550/arXiv.2509.09821](https://arxiv.org/abs/10.48550/arXiv.2509.09821)
- 2026 *Kuan-Fu Feng*, **Marine Denolle**, Fan-Chi Lin, Tonie van Dam (2026). **A Decadal Survey of the Near-Surface Seismic Velocity Response to Hydrological Variations in Utah, United States.** *Journal of Geophysical Research: Solid Earth*, 131
- 2025 *Akash Kharita*, **Marine Denolle**, Alexander R Hutko, J Renate Hartog, Stephen D Malone (2025). **Exploration of Machine Learning Methods to Seismic Event Discrimination in the Pacific Northwest.** accepted in *Seismica*, pre-print on Arxiv, [10.48550/arXiv.2510.23795](https://arxiv.org/abs/10.48550/arXiv.2510.23795)
- 2025 *Yiyu Ni*, **Marine Denolle**, Amanda Thomas, Alex Hamilton, Jannes Münchmeyer, Yinzhi Wang, Loïc Bachelot, Chad Trabant, David Mencin (2025). **A Global-scale Database of Seismic Phases from Cloud-based Picking at Petabyte Scale.** *Seismica*, 4, [10.26443/seismica.v4i2.1738](https://doi.org/10.26443/seismica.v4i2.1738)
Featured in: [Video](#)
- 2025 *Yiyu Ni*, **Marine A Denolle**, Jannes Münchmeyer, Yinzhi Wang, *Kuan-Fu Feng*, Carlos Garcia Jurado Suarez, Amanda M Thomas, Chad Trabant, Alex Hamilton, David Mencin (2025). **A Review of Cloud Computing and Storage in Seismology.** *Geophysical Journal International*, [10.1093/gji/ggaf322](https://doi.org/10.1093/gji/ggaf322)

- 2025 **Marine A. Denolle**, Carl Tape, Ebru Bozdağ, Yinzhi Wang, Felix Waldhauser, Alice-Agnes Gabriel, Jochen Braunmiller, Bryant Chow, Liang Ding, *Kuan-Fu Feng*, Ayon Ghosh, Nathan Groebner, Aakash Gupta, *Zoe Krauss*, Amanda M. McPherson, Masaru Nagaso, Zihua Niu, *Yiyu Ni*, Rıdvan Örsvuran, Gary Pavlis, Felix Rodriguez-Cardozo, Theresa Sawi, David Schaff, Nico Schliwa, David Schneller, *Qibin Shi*, Julien Thurin, Chenxiao Wang, Kaiwen Wang, Jeremy Wing Ching Wong, Sebastian Wolf, *Congcong Yuan* (2025). **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing Through Diverse Training Modalities**. *Seismological Research Letters*, 96, [10.1785/0220240413](https://doi.org/10.1785/0220240413)
- 2025 **Marine A. Denolle**, *Qibin Shi*, *Tim Clements*, *Loïc Viens*, Veronica Rodriguez-Tribaldos, Fabrice Cotton (2025). **Ambient field seismology in critical zone hydrological sciences**. *Comptes Rendus. Géoscience*, 357, [10.5802/crgeos.310](https://doi.org/10.5802/crgeos.310)
- 2025 *Q. Shi*, **M. A. Denolle**, *Y. Ni*, *E. F. Williams*, N. You (2025). **Denoising Offshore Distributed Acoustic Sensing Using Masked Auto-Encoders to Enhance Earthquake Detection**. *JGR: Solid Earth*, 130, [10.1029/2024JB029728](https://doi.org/10.1029/2024JB029728)
- 2025 *Q. Shi*, *E. F. Williams*, B. P. Lipovsky, **M. A. Denolle**, W. S. D. Wilcock, D. S. Kelley, K. Schoedl (2025). **Multiplexed Distributed Acoustic Sensing Offshore Central Oregon**. *Seismological Research Letters*, 96, [10.1785/0220240460](https://doi.org/10.1785/0220240460)
Featured in: [UW News](#), [Phys](#)
- 2024 *Y. Ni*, **M. A. Denolle**, *Q. Shi*, B. P. Lipovsky, S. Pan, J. N. Kutz (2024). **Wavefield reconstruction of distributed acoustic sensing: Lossy compression, wavefield separation, and edge computing**. *Journal of Geophysical Research: Machine Learning and Computation*, 1, [10.1029/2024JH000247](https://doi.org/10.1029/2024JH000247)
- 2024 P. Makus, **M. A. Denolle**, C. Sens-Schönfelder, *M. Köpfli*, F. Tilmann (2024). **Analysing Volcanic, Tectonic, and Environmental Influences on the Seismic Velocity from 25 Years of Data at Mount St. Helens**. *Seismological Research Letters*, 95, [10.1785/0220240088](https://doi.org/10.1785/0220240088)
- 2024 M. Köpfli, **M. A. Denolle**, W. Thelen, P. Makus, S. Malone (2024). **Examining 22 Years of Ambient Seismic Wavefield at Mount St. Helens**. *Seismological Research Letters*, 95, [10.1785/0220240079](https://doi.org/10.1785/0220240079)
- 2024 F. Diewald, **M. Denolle**, J. J. Timothy, C. Gehlen (2024). **Impact of Temperature and Relative Humidity Variations on Coda Waves in Concrete**. *Scientific Reports*, 14, [10.1038/s41598-024-69564-4](https://doi.org/10.1038/s41598-024-69564-4)
- 2024 *K. Okubo*, B. Delbridge, **M. Denolle** (2024). **Monitoring velocity change over 20 years at Parkfield**. *Journal of Geophysical Research: Solid Earth*, 129, [10.1029/2023JB028084](https://doi.org/10.1029/2023JB028084)
- 2024 T. Cochard, I. Svetlizky, G. Albertini, R. C. Viesca, S. M. Rubinstein, F. Spaepen, *C. Yuan*, **M. Denolle**, Y.-Q. Song, L. Xiao, D. A. Weitz (2024). **Extended crack propagation by local nucleation and rapid transverse expansion**. *Nature Physics*, [10.1038/s41567-023-02365-0](https://doi.org/10.1038/s41567-023-02365-0)
- 2023 *A. Kharita*, **M. Denolle**, M. West (2023). **Discrimination between icequakes and earthquakes in southern Alaska: an exploration of waveform features using random forest algorithm**. *Geophysical Journal International*, [10.1093/gji/ggae106](https://doi.org/10.1093/gji/ggae106)
- 2023 *S. Olinger*, B. Lipovsky, **M. Denolle** (2023). **Ocean Coupling Limits Rupture Velocity of Fastest Observed Ice Shelf Rift Propagation Event**. *AGU Advances*, 5, [10.1029/2023AV001023](https://doi.org/10.1029/2023AV001023)
Featured in: [UW News](#), [EOS](#)
- 2023 *C. Yuan*, T. Cochard, **M. Denolle**, J. Gomberg, A. Wech, L. Xiao, D. Weitz (2023). **Laboratory hydrofracture as analogs to tectonic tremors**. *AGU Advances*, 5, [10.1029/2023AV001002](https://doi.org/10.1029/2023AV001002)
Featured in: [UW News](#), [EOS](#)
- 2023 *Q. Shi*, **M. Denolle** (2023). **Improved observations of deep earthquake ruptures using machine learning**. *Journal of Geophysical Research: Solid Earth*, 128, [10.1029/2023JB027334](https://doi.org/10.1029/2023JB027334)
- 2023 *Congcong Yuan*, *Yiyu Ni*, Youzuo Lin, **Marine Denolle** (2023). **Better Together: Ensemble Learning for Earthquake Detection and Phase Picking**. *IEEE Transactions on Geoscience and Remote Sensing*, 61, [10.1109/TGRS.2023.3320148](https://doi.org/10.1109/TGRS.2023.3320148)

- 2023 Yiyu Ni, **Marine A. Denolle**, Rob Fatland, Naomi Alterman, Bradley P. Lipovsky, Friedrich Knuth (2023). **An Object Storage for Distributed Acoustic Sensing**. Seismological Research Letters, 95, [10.1785/0220230172](https://doi.org/10.1785/0220230172)
- 2023 Z. Krauss, Y. Ni, S. Henderson, **M. Denolle** (2023). **Seismology in the cloud: guidance for the individual researcher**. Seismica, 2, [10.26443/seismica.v2i2.979](https://doi.org/10.26443/seismica.v2i2.979)
- 2023 Y. Ni, A. Hutko, F. Skene, **M. Denolle**, S. Malone, P. Bodin, R. Hartog, A. Wright (2023). **Curated Pacific Northwest AI-ready Seismic Dataset**. Seismica, 2, [10.26443/seismica.v2i1.368](https://doi.org/10.26443/seismica.v2i1.368)
- 2023 L. Ermert, E. Cabral-Cano, E. Chaussard, D. Solano-Rojas, L. Quintanar, D. Morales Padilla, E. A. Fernandez-Torres, **M. A. Denolle** (2023). **Probing environmental and tectonic changes underneath Ciudad de México with the urban seismic field**. Solid Earth (EGU), [10.5194/egusphere-2022-1361](https://doi.org/10.5194/egusphere-2022-1361)
- 2023 T. Clements, **M. A. Denolle** (2023). **The Seismic Signature of California's Earthquakes, Droughts, and Floods**. Journal of Geophysical Research: Solid Earth, 128, [10.1029/2022JB025553](https://doi.org/10.1029/2022JB025553)
- 2022 J. Yin, **M. A. Denolle**, B. He (2022). **A multitask encoder–decoder to separate earthquake and ambient noise signal in seismograms**. Geophysical Journal International, 231, [10.1093/gji/ggac290](https://doi.org/10.1093/gji/ggac290)
- 2022 C. Jiang, **M. A. Denolle** (2022). **Pronounced Seismic Anisotropy in Kanto Sedimentary Basin: A Case Study of Using Dense Arrays, Ambient Noise Seismology, and Multi-Modal Surface-Wave Imaging**. Journal of Geophysical Research: Solid Earth, 127, [10.1029/2022JB024613](https://doi.org/10.1029/2022JB024613)
- 2022 L. Viens, C. Jiang, **M. A. Denolle** (2022). **Imaging the Kanto Basin seismic basement with earthquake and noise autocorrelation functions**. Geophysical Journal International, 230, [10.1093/gji/ggac101](https://doi.org/10.1093/gji/ggac101)
- 2022 S. D. Olinger, B. P. Lipovsky, **M. A. Denolle**, B. W. Crowell (2022). **Tracking the Cracking: a Holistic Analysis of Rapid Ice Shelf Fracture Using Seismology, Geodesy, and Satellite Imagery on the Pine Island Glacier Ice Shelf, West Antarctica**. Geophysical Research Letters, [10.1029/2021GL097604](https://doi.org/10.1029/2021GL097604)
- 2022 Z. Yang, C. Yuan, **M. A. Denolle** (2022). **Detecting Elevated Pore Pressure due to Wastewater Injection Using Ambient Noise Monitoring**. The Seismic Record, 2, [10.1785/0320210036](https://doi.org/10.1785/0320210036)
- 2021 J. Yin, **M. A. Denolle** (2021). **The Earth's Surface Controls the Depth-Dependent Seismic Radiation of Megathrust Earthquakes**. AGU Advances, 2, [10.1029/2021AV000413](https://doi.org/10.1029/2021AV000413)
Featured in: [EOS](https://www.eos.org)
- 2021 C. Yuan, J. Bryan, **M. A. Denolle** (2021). **Comparing approaches to measuring seismic phase variations in the time, frequency, and wavelet domains**. Geophysical Journal International, 226, [10.1093/gji/ggab140](https://doi.org/10.1093/gji/ggab140)
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Grants

- 2025 **Jerry and Linda Paros** Mult-Sensor, Multi-Geohazards Monitoring at Mt Rainier and Mountaineous regions: \$750,000 (lead PI)
- 2025 **UW CRESST - FFST** Toward Predictive Understanding of Climate-Compounded Geohazards: \$300,000 (lead PI)
- 2025 **NSF CS4AI** Separating the Signal from the Noise: Promoting Alaskan students inquiry with geographically relevant seismic data and machine learning techniques: \$889,766 (lead PI Lore, Denolle co-PI, \$62,000 to UW), DRL-2524060
- 2025 **NSF RISE CAIG** Collaborative Research: CAIG: Framework for Artificial Intelligence-Enhanced Modeling of Wildfire Geohazards (FAIM-WG): Applications for postfire debris flows across the Western US: \$735,074 (lead PI Istanbuluoglu, Denolle co-PI), RISE-2530591
- 2025 **NSF OCE** Multi-span distributed fiber sensing on the Ocean Observatories Initiative Regional Cabled Array: \$880,490 (lead PI Wilcock, Denolle co-PI), OCE-2526198, technology development grant.
- 2025 **NSF OCE** EXTension on the ENDeavour Segment (EXTEND): Illuminating the seafloor spreading cycle: \$785,819.00 (lead PI Wilcock, Denolle co-PI), OCE-2439442
- 2024 **NSF EAR** Collaborative Research: A seismic investigation of slow slip and fault locking along the Alaska-Aleutian subduction zone: \$248,835.00 (Denolle co-PI, collaborative with lead PI Golos), EAR-2346079
- 2023 **NSF OCE** RAPID: Multiplexed Distributed Acoustic Sensing (DAS) at the Ocean Observatory Initiative (OOI) Regional Cabled Array (RCA): \$198,069.00 (lead PI Wilcock, Denolle co-PI), OCE-2415521
- 2023 **Ecole Normal Supérieure de Paris** Travel Fellowship: \$3,500 (Denolle PI)
- 2023 **SCEC** CyberTraining for Seismology: Data Science and HPC: \$35,229 (Denolle PI)
- 2021 **David and Lucile Packard Foundation** URG Program: \$50,000 (PI)

- 2021 **Murdock Charitable Trust Fund** UW FiberLab equipment: \$950,000 (co-PI, lead PI Brad Lipovsky). Built facility for Fiber Sensing.
- 2021 **NSF CyberTraining** GeoSMART ML workforce development: \$995,817 (co-PI). Build a course MLGEO.
- 2021 **NSF OAC-CSSI** Collaborative Research: Frameworks: Seismic COmputational Platform for Empowering Discovery (SCOPED): \$717,441.00 (Denolle PI, lead-PI Carl Tape), OAC-2103701
- 2021 **NSF EAR, PREEVENTS** Collaborative Research: Cross-Validation of Empirical and Physics-based ground motion predictions: \$167,804.00 (Denolle PI), EAR-2125337
- 2020 **SCEC grant** Aftershock patterns and co-seismic off-fault damage elucidate dynamic rupture processes on the 2019 Ridgecrest earthquake sequence: \$33,307 (Denolle PI) - returned
- 2019 **Harvard University - David Rockefeller Center for Latin American Studies** Monitoring Seismic Hazards in Mexico City using Grillo, a Low-Cost Earthquake Early Warning System, Hardware, \$85,000 (Denolle PI)
- 2019 **Harvard Data Science Initiative** Ambient-noise seismology using Cloud Computing: \$27,210 (Denolle PI)
- 2018 **NSF EAR, CAREER** CAREER: Dynamics of surface rupturing thrust earthquakes: \$504,315 (PI)
- 2018 **SCEC grant** Data Collection for Virtual Earthquakes on Cajon Pass: \$28,085 (Denolle PI) - field work
- 2017 **David and Lucile Packard Foundation** Fellowship: \$875,000 (PI)
- 2017 **NSF PREEVENTS** Collaborative Proposal - PREEVENTS Track 2: Cascadia Scenario Earthquakes: Source, Path, and implications for Earthquake Early Warning: \$324,495.00 (Denolle PI, collaborative with lead PI Yihe Huang and Amanda Thomas), EAR-1739712
- 2017 **SCEC grant** Static and dynamic source parameters of global strike-slip earthquakes: \$25,000 (Denolle PI)
- 2016 **SCEC grant** Epistemic uncertainties in ground motion prediction from virtual earthquakes: \$26,173 (Denolle PI)
- 2015 **SCEC grant** Basin Response to Virtual Earthquakes on the San Jacinto Fault and the Itoigawa-Shizuoka Fault: \$20,000 (Denolle PI)

Research Products

- 2021 – present **Machine Learning in the Geosciences (Open-Access Jupyter Book)** Textbook
[Book](#) and [course repository](#) provide graduate-level ML curriculum, asynchronous modules, and home-work data pipelines for geoscience training.
- Addresses a core gap: no canonical ML textbook existed for geoscience graduate classrooms.
 - Adoption and contribution interest from peer programs (including Arizona and Berkeley).
 - Ongoing community development with curated datasets and reproducible teaching workflows.
- 2019 – present **NoisePy** Software
 Open-source Python software for large-scale ambient-noise seismology. [GitHub](#).
- Community scale: 207 stars, 83 forks, 19 contributors (Jan 2026).
 - Operational usage: taught in workshops and used for large-scale cross-correlation workflows.
 - Open, version-controlled development supporting reproducible and cloud-oriented processing.

- 2020 – present **SeisNoise.jl** Software
Open-source Julia software for high-performance ambient-noise processing and cross-correlation. [GitHub](#).
- Extends the Julia seismology ecosystem with core ambient-noise tooling.
 - Community signal: 50 stars and 17 forks (Apr 2023 snapshot).
 - Used by group members for production research workflows.
- 2023 **EarthML4PNW / PNW AI-Ready Seismic Dataset** Dataset
Curated ML-ready Pacific Northwest dataset product. [Dataset repository](#).
- Publication link: Seismica (2023), DOI [10.26443/seismica.v2i1.368](#).
 - Version-controlled release and metadata stewardship through GitHub workflows.
 - Designed for reproducible benchmarking and graduate-level ML coursework.
- 2025 **Global Seismic Phase Database** Dataset
Global seismic data product from cloud-scale waveform processing. [DOI](#).
- Scale: 4.3 billion phase picks from 1.3 PB of continuous seismic data.
 - Publication link: Seismica (2025) with DOI-backed archival record.
 - Supports global monitoring and data-driven geophysics workflows.

Invited Talks

- 2026 **NGF Science Meeting** Conference Speaker
- 2026 **WCCM 2026, Munich** Conference Speaker
- 2026 **European Geophysical Union** Conference Speaker
- 2026 **CERI Memphis** Department Colloquium
- 2025 **SIAM Geoscience Meeting** Plenary Speaker
- 2025 **Workshop on Earthquake Physics and Applications of AI to Tectonic Faulting, Italy** Plenary Speaker
- 2025 **Civil Environmental Engineering, University of Washington** Department Colloquium
- 2025 **Radcliffe Institute of Advanced Studies, on the road** Short Talk
- 2025 **Washington University, Saint Louis** Department Colloquium
- 2024 **University of Southern California** Department Colloquium
- 2024 **University of California, Davis** Department Colloquium
- 2024 **Northern Arizona University** Department Colloquium
- 2024 **Passive imaging and monitoring in wave physics (Cargese, France)** Invited Conference Talk
- 2024 **Statewide California Earthquake Center** Plenary Speaker
- 2023 **Ecole Normale Supérieure, Paris Séminaire** Departemental
- 2023 **eScience Institute, University of Washington** Data Science Seminar

2023 **Sandia National Lab, GNEM seminar series** Department Colloquium

2022 **American Geophysical Union (x2)** Invited Conference Talk

2022 **University of New Mexico** Department Colloquium

2021 **University of Oregon** Seismo Colloquium

2021 **U Utah, Seismo Tea** Seismo Colloquium

2021 **University of Wisconsin** Department Colloquium

2021 **Colorado School of Mines** Department Colloquium

2021 **Mexico a traves de los sismos** Invited Conference Talk

2020 **University of Washington** Department Colloquium

2020 **UC Berkeley** Department Colloquium

2019 **Yale University** Department Colloquium

2019 **University of Washington, seismolunch** Seismo Colloquium

2019 **EGU annual meeting, Vienna** Invited Conference Talk

2019 **Michigan State University** Department Colloquium

2019 **Victoria University of Wellington, SN Jepson Lecture, New Zealand** Public Lecture

2019 **GNS-Science, New Zealand** Department Colloquium

2019 **Stanford University, Department of Geophysics** Department Colloquium

2019 **Tufts University, Civil Engineering seminar** Department Colloquium

2018 **Brown University** Department Colloquium

2018 **Ecole Normale Supérieure, Paris** Department Colloquium

2017 **AGU, New Orleans** Invited Conference Talk

2017 **Columbia University, Lamont Doherty Earth Observatory** Department Colloquium

2016 **Harvard Museum of Natural History** Public Lecture

2016 **University of Oregon** Department Colloquium

2016 **University of New Hampshire, Chapman Colloquium** Department Colloquium

2016 **UC Santa Cruz, IGPP seminar** Department Colloquium

2016 **Massachusetts Institute of Technology** Department Colloquium

2015 **USGS, Menlo Park, Earthquake Hazard Program** Department Colloquium

- 2015 **University of Victoria, BC, Canada** Department Colloquium
- 2015 **Penn State, Geodynamics seminar** Department Colloquium
- 2015 **Harvard, Earth and Planetary Sciences** Department Colloquium
- 2015 **UT Austin, Solid Earth seminar** Department Colloquium
- 2015 **UCLA, seismology/tectonics seminar** Department Colloquium
- 2015 **HOKUDAN International Symposium on Active Faulting, Awaji, Japan** Invited Conference Talk
- 2015 **Information Theory and Applications workshop, La Jolla** Invited Conference Talk
- 2015 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium
- 2015 **University of Southern California, Geophysics seminar** Department Colloquium
- 2014 **Strong Motion Studies for Future Mega-Quakes, DPRI, Kyoto University, Japan** Invited Conference Talk
- 2014 **AGU-SEG Summer Research workshop, Vancouver, Canada** Invited Conference Talk
- 2014 **San Diego State University** Department Colloquium
- 2014 **UC Santa Barbara** Department Colloquium
- 2014 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium
- 2013 **AGU, Meeting of the Americas, Cancun, Mexico** Invited Conference Talk
- 2013 **Caltech Seismo Lab** Department Colloquium
- 2013 **USGS, Menlo Park** Department Colloquium
- 2013 **Stanford ICME seminar** Department Colloquium
- 2013 **Earthquake Research Institute, Tokyo University, Japan** Department Colloquium
- 2013 **Advanced Industrial Science and Technology, Japan** Department Colloquium
- 2013 **Disaster Prevention Research Institute, Japan** Department Colloquium
- 2012 **ACOUSTICS, France** Invited Conference Talk
- 2012 **Berkeley Seismo Lab** Department Colloquium
- 2011 **Institut de Physique du Globe de Paris, Earthquake seminar** Department Colloquium